Purchasing Request

Electronics Assembly for a 1030-nm Fiber Laser Lidar Transmitter

Statement of Work (SOW)

Background

NASA GSFC is developing a dual wavelength small all-range lidar (SALI-DW) for lunar science and exploration under a GSFC FY26 IRAD funding. One of the major tasks is to develop a prototype fiber laser transmitter at 1030 nm wavelength and 2-W output power. The circuit design must be based on an existing circuit which has demonstrated a Technology Readiness Level 6 (TRL-6) for space application.

Specifications

Fabricate, test, and deliver a prototype electronics circuit assembly for a 1030-nm 2-W fiber laser transmitter with direct heritage from the GSFC Hazard Detection Lidar or a similar space lidar which has demonstrated TRL-6 through testing.

- Laser output power: 2 W average, 256 W peak.
- Laser output wavelength: 1030 nm.
- Modulation: Return to Zero Pseudo Noise (RZPN) code with 8-ns pulse width and 1/128 duty cycle.
- Output power adjustment range: continuous adjustment from 0 to 2-W average output power.
- Input/Power: 28 VDC.
- Input/Signal: Laser trigger, 0 to 1 V, 8-ns pulse width.
- Digital Input/Output: laser control and telemetry read-back.
- Operation temperature: 0 to 50 degrees Celsius.
- Operation condition (pressure): <1e-5 torr.
- Size: 184.15 by 101.6 mm printed circuit board, one each.

Period of Performance

Four months after receipt of order (ARO).

Deliverables

(1) Electronics circuit assembly for a 1030-nm 2-W fiber laser lidar transmitter, quantity one.